Circles and Parabolas in Standard Form

Completing the square to put a circle equation into centre-radius form and a quadratic into vertex form, and then reading the graph off the finished form

How to use this sheet.

- **Circle, standard form:** $(x-h)^2+(y-k)^2=r^2$, centre (h,k), radius r. **Parabola, vertex form:** $y=a(x-h)^2+k$, vertex (h,k).
- Completing the square is the same move both times: take half the coefficient of the linear term, square it, and add that number — balancing the equation as you go.
- Group the x terms together and the y terms together first, and move the constant out of the way before you start.
- Read the signs off the finished form carefully. $(x+4)^2$ means $h=-4$, and the number on the right of a circle equation is r^2, not r.

1. The completing move on its own. Fill in the number that makes the left side a perfect square, then write it as a squared binomial.

$x^2 - 10x + \_\_\_\_\_\_ \;=\; \left(x - \_\_\_\_\_\right)^2$

Number added: ________ **Squared binomial:** __________________

2. A circle from general form. Rewrite in standard form, then state the centre and the radius.

$$x^2 + y^2 - 6x + 4y - 12 = 0$$

Standard form: ________________________________

Centre: (________, ________) **Radius:** ________

3. Watch the signs. Rewrite in standard form, then state the centre and the radius.

$$x^2 + y^2 + 8x - 2y + 8 = 0$$

Standard form: ________________________________

Centre: (________, ________) **Radius:** ________

4. A parabola into vertex form. Rewrite in vertex form, then state the vertex.

$$y = x^2 - 8x + 11$$

Vertex form: ________________________________

Vertex: (________, ________)

5. Vertex form and the x-intercepts. Rewrite in vertex form and state the vertex, then solve $y = 0$ to find where the parabola crosses the x-axis.

$$y = x^2 + 6x + 5$$

Vertex form: ______________________________

Vertex: (________, ________) **x-intercepts:** $x =$ ________ and $x =$ ________

6. A circle with a coefficient in front. The squared terms do not have coefficient 1, so deal with that before completing the square. Rewrite in standard form and state the centre and radius.

$$2x^2 + 2y^2 - 8x + 12y - 24 = 0$$

Standard form: _________________________________

Centre: (________, ________) **Radius:** ________

7. A parabola with a coefficient in front. Factor before you complete the square. Rewrite in vertex form and state the vertex.

$$y = 2x^2 - 12x + 13$$

Vertex form: ________________________________

Vertex: (________, ________)

8. Inside, on, or outside? Rewrite the circle in standard form. Then work out how far the point $(-2, 1)$ is from the centre, compare that distance with the radius, and say where the point lies.

$$x^2 + y^2 + 10x - 6y + 18 = 0$$

Standard form: _____________________________

Distance to centre: ________ **Radius:** ________ **The point is** ___________

9. Back the other way. This circle is already in standard form. Expand and collect terms to write it in general form, with 0 on the right.

$$(x+1)^2 + (y-4)^2 = 20$$

General form: ______________________________ $= 0$

10. A parabola on its side. This one is solved for x, so the square gets completed in y. Rewrite it in vertex form and state the vertex.

$$x = y^2 - 4y + 7$$

Vertex form: ________________________________

Vertex: (________, ________) the vertex is (x, y), so read the two numbers in the right order

11. Where the line meets the circle. Substitute the line into the circle, solve the quadratic that results, and then find the matching y for each x.

$$x^2 + y^2 = 25 \qquad \text{and} \qquad y = x + 1$$

Intersection points: (_________, _________) and (_________, _________)

12. A circle that isn't there.

$$x^2 + y^2 - 4x + 6y + 20 = 0$$

(a) Complete the square on both variables and write what the equation becomes.

Completed form: ______________________________

(b) What does that result tell you about the graph of this equation? Explain in two or three sentences, and say why.